Teaching Generalizable Cooperation in Multi-Agent Reinforcement Learning

Similoluwa Okunowo
African Institute for Mathematical Sciences (AIMS) South Africa
`similoluwa@aims.ac.za`

Abstract

We present an open-source, multimodal educational resource for generalizable cooperation in multi-agent reinforcement learning (MARL): how agents coordinate, communicate, and adapt when partners, communication conventions, or deployment conditions differ from training. The resource uses structured, bite-sized lessons, animated explainers, interactive practice, virtual and practical Jupyter notebooks, to move learners from cooperative MARL foundations to advanced topics like cross-play, unseen-partner evaluation, ad hoc teamwork, and zero-shot coordination. It culminates in the emerging frontier of learned collaboration among LLM agents, where the same questions of coordination and adaptation reappear.

Tutorial website: rexsimiloluwah.github.io/marl-for-cooperative-environments

1 Concept

In multi-agent reinforcement learning (MARL), where multiple agents learn to act toward a shared goal, **generalizable cooperation** asks whether effective teamwork can persist when partners, communication conventions, or deployment conditions differ from training. This matters in robotics, autonomous vehicles, network optimization, and resource allocation, and increasingly in learned collaboration among LLM agents.

This resource teaches MARL in cooperative environments through a guided learning journey rather than as a set of fragmented topics. It begins with single-agent reinforcement learning, builds through coordination and communication, then reaches partner adaptation, zero-shot teamwork, and the frontier of LLMs as cooperative agents.

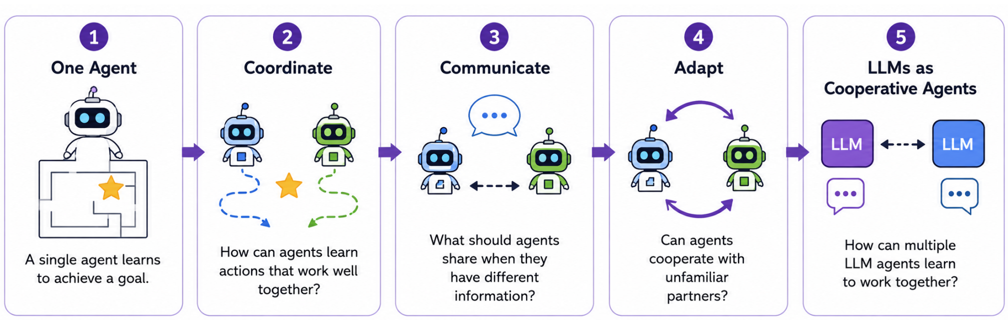

Figure 1: The five stages of the journey. Each challenge is created by the one before it.

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2 Audience and Prerequisites

Advanced undergraduate and graduate learners with basic reinforcement learning, Python, probability, and linear algebra; no prior MARL required. The textbook by Albrecht et al. [1] is a useful companion for broader coverage.

3 Learning Objectives

The resource follows Bloom's Taxonomy, progressing from understanding core concepts to applying, analysing, evaluating, and creating cooperative multi-agent systems.

- **Understand:** explain coordination, communication, credit assignment, CTDE, and partner dependence.
- **Apply:** use and compare cooperative learning approaches in small environments.
- **Analyse and Evaluate:** diagnose failures under communication limits and changing partners.
- **Create:** design and justify a cooperative MARL system that generalizes beyond its training partners.

4 Teaching Materials

The resource uses a multimodal learning design, combining structured explanation, visual storytelling, retrieval practice, and hands-on experimentation.

Teaching material	What it provides
Interactive tutorial website	Structured, bite-sized lessons with intuitive explanations, equations, research connections, flashcards, embedded knowledge checks, and interactive worksheets.
Explainer videos	Two animated explainers, on the core MARL journey (youtu.be/TOJrFLJHXkM) and on LLMs as cooperative agents (youtu.be/P5OAe5XLmzc).
Labs	In-browser virtual labs and Colab notebooks for hands-on practice with coordination, communication, partner generalization, and wireless network resource allocation.
`cooperative-marl-labs`	Open-source Python package with reusable environments, agents, partner policies, training utilities, evaluation tools, and visualizations.

All teaching materials and source code are available in the GitHub repository:
github.com/rexsimiloluwah/marl-for-cooperative-environments

5 Linked Recent Papers

The learning path is grounded in recent work from 2022 to 2026 that demonstrates the concept is active at the research frontier.

Learning question	Recent papers using the concept
Evaluating coordination	SMACv2 [2], NeurIPS Datasets and Benchmarks 2023.
Learned communication	LangGround [3], NeurIPS 2024.
Unfamiliar partners	N-Agent Ad Hoc Teamwork [4], NeurIPS 2024; ZSC-Eval [5], NeurIPS Datasets and Benchmarks 2024.
LLM collaboration	MAPoRL [6], ACL 2025; LLM Collaboration with MARL [7], AAAI 2026.

References

[1] S. V. Albrecht, F. Christianos, and L. Schäfer. *Multi-Agent Reinforcement Learning: Foundations and Modern Approaches*. MIT Press, 2024. marl-book.com.

[2] B. Ellis et al. SMACv2: an improved benchmark for cooperative MARL. *NeurIPS Datasets and Benchmarks*, 2023.

[3] H. Li et al. Language grounded MARL with human-interpretable communication. *NeurIPS*, 2024.

[4] C. Wang et al. N-agent ad hoc teamwork. *NeurIPS*, 2024.

[5] X. Wang et al. ZSC-Eval: a benchmark for multi-agent zero-shot coordination. *NeurIPS Datasets and Benchmarks*, 2024.

[6] C. Park et al. MAPoRL: multi-agent post-co-training for collaborative LLMs with RL. *ACL*, 2025.

[7] S. Liu et al. LLM collaboration with multi-agent reinforcement learning. *AAAI*, 2026.

AI Usage and Educational Integrity. Generative AI tools supported parts of the writing, coding, visual design, and iteration of the resource. All educational content and technical claims were reviewed before inclusion. Simplified environments in the Python package are designed for teaching and experimentation, **not for evaluating or comparing MARL algorithms for scientific claims**. The resource is **actively maintained and will continue to be updated as the field evolves and learner or instructor feedback is incorporated**.